

PEECHARA SATHWIK

Python Developer | AI & Data Analytics

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SUMMARY

Aspiring Python Developer with hands-on experience in NLP, Computer Vision, and AI-powered application development. Skilled in building end-to-end ML pipelines and integrating large language model APIs. Experienced in backend development, debugging, and developing scalable AI solutions.

EDUCATION

B.Tech in Computer Science (Data Science), Malla Reddy Engineering College, Hyderabad | 2023-2027
•**CGPA: 7.47 / 10**
Senior Secondary (XII), TSBIE - Science, Alphores Junior College Karimnagar | 2021-2023
•**Percentage: 70.4%**
Secondary (X), TSBSE, Alphores High School Karimnagar | 2021
•**CGPA: 10.00 / 10**

SKILLS

- **Languages:** Python, Java
- **Databases:** SQL: MySQL (RDBMS); familiar with NoSQL concepts
- **Cloud:** Google Cloud Fundamentals
- **AI / ML:** NLP, Transformers, Computer Vision, Machine Learning, Generative AI, Stable Diffusion, LoRA Fine-Tuning, Prompt Engineering
- **Libraries:** NumPy, Pandas, OpenCV, Pillow, Diffusers, Hugging Face Transformers
- **Tools:** Git, GitHub, Kaggle (cloud-based GPU training)

PROJECTS

NexusAI – AI Chatbot with NLP & Multimodal Input *Python | Flask | Groq API | Google Gemini API | MySQL | SQLAlchemy | NLP | PyMuPDF* 2026

- Full-stack AI chatbot: user input → Flask /chat route → Groq/Gemini (auto-failover) → real-time NLP analysis (intent/sentiment/language/entities) → MySQL logging (SQLAlchemy) → /analytics dashboard for usage trends.
- Challenge: rule-based NLP misfired on sarcasm and code-switched (Telugu/Hindi/English) input due to no contextual understanding; documented as a limitation needing a multilingual model (e.g. XLM-RoBERTa) for production. Link: https://github.com/sathwik2021/nlp_chatbot_enhanced

AI Manga Generation Pipeline – Text-to-Manga System *Python | Flask | Stable Diffusion 1.5 | LoRA | Gemini API | OpenCV | Pillow | Hugging Face* 2025 – 2026

- Text-to-manga system: raw story text → Gemini (narrative beat extraction) → Gemini (20-rule panel layout engine) → LoRA-fine-tuned Stable Diffusion 1.5 (panel image synthesis) → OpenCV/Pillow (compositing into final page) → Flask web UI with live progress.
- Challenge: rule-based NLP misfired on sarcasm and code-switched (Telugu/Hindi/English) input due to no contextual understanding; documented as a limitation needing a multilingual model (e.g. XLM-RoBERTa) for production.

Codex Council – Multi-Agent LLM Code Review Pipeline *Python | Ollama | Groq / Gemini / Mistral / OpenRouter APIs* 2026

- Multi-agent code-review tool: codebase → indexed (BM25 + vector embeddings) → Planner routes query to specialist teams (security, architecture, bugs, optimization) → each team runs Generator→Critic→Verifier debate → Mediator resolves cross-team conflicts → final report.
- Challenge: LLM-generated JSON outputs broke on embedded multiline code; fixed after three iterations by anchoring extraction on the schema's next known field name instead of regex.

CERTIFICATIONS

- Azure AI Fundamentals (AI-900) — Microsoft
- Data Analytics Job Simulation — Deloitte (Forage)
- Machine Learning Certification — Tutedude (Nov 2025)
- Internship Certificate — AimR Edu (2024)
- Data Structures — TechAugusta (Oct 2024)
- Google Cloud Computing Foundations — Google (Apr 2026)